Experiment 4: Statistical Analysis, Visualization, and Outlier Detection

---

Aim

To perform basic statistical analysis, visualize the dataset using plots, and detect outliers using the Interquartile Range (IQR) method.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("/content/titanic - titanic.csv")

# ------------------------------
# 1️⃣ BASIC STATISTICAL ANALYSIS
# ------------------------------

print("Dataset Info:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values Count:")
print(df.isnull().sum())

# ------------------------------
# 2️⃣ DATA VISUALIZATION
# ------------------------------

# Histogram - Age
plt.hist(df['Age'].dropna(), bins=10)
plt.xlabel("Age")
plt.ylabel("Count")
plt.title("Age Distribution")
plt.show()

# Histogram - Fare
plt.hist(df['Fare'].dropna(), bins=10)
plt.xlabel("Fare")
plt.ylabel("Count")
plt.title("Fare Distribution")
plt.show()

# Survival count bar plot
df['Survived'].value_counts().plot(kind='bar')
plt.xlabel("Survived (0 = No, 1 = Yes)")
```

```
plt.ylabel("Number of Passengers")
plt.title("Survival Count")
plt.show()

# Boxplot for Age and Fare
df[['Age', 'Fare']].plot(kind='box')
plt.title("Boxplot for Age and Fare")
plt.show()

# -------------------------------
# 3️⃣ OUTLIER ANALYSIS (IQR METHOD)
# -------------------------------

Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Fare'] < lower_bound) | (df['Fare'] > upper_bound)]

print("\nFare Outliers:")
print(outliers[['Fare']].head())

print("\nNumber of Fare Outliers:", len(outliers))
```

Output

```
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
```

10 Cabin   204 non-null   object
11 Embarked   889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

Statistical Summary:

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Missing Values Count:

PassengerId   0
Survived   0
Pclass   0
Name   0

Sex 0

Age 177

SibSp 0

Parch 0

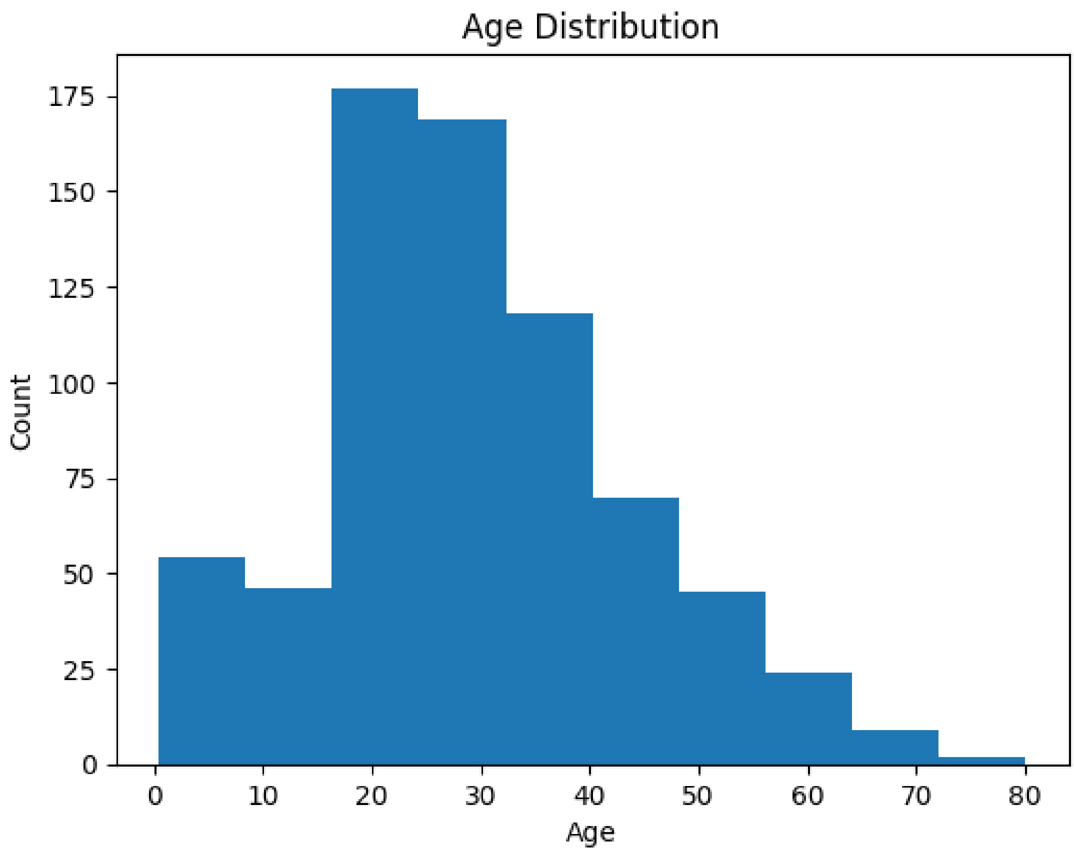
Ticket 0

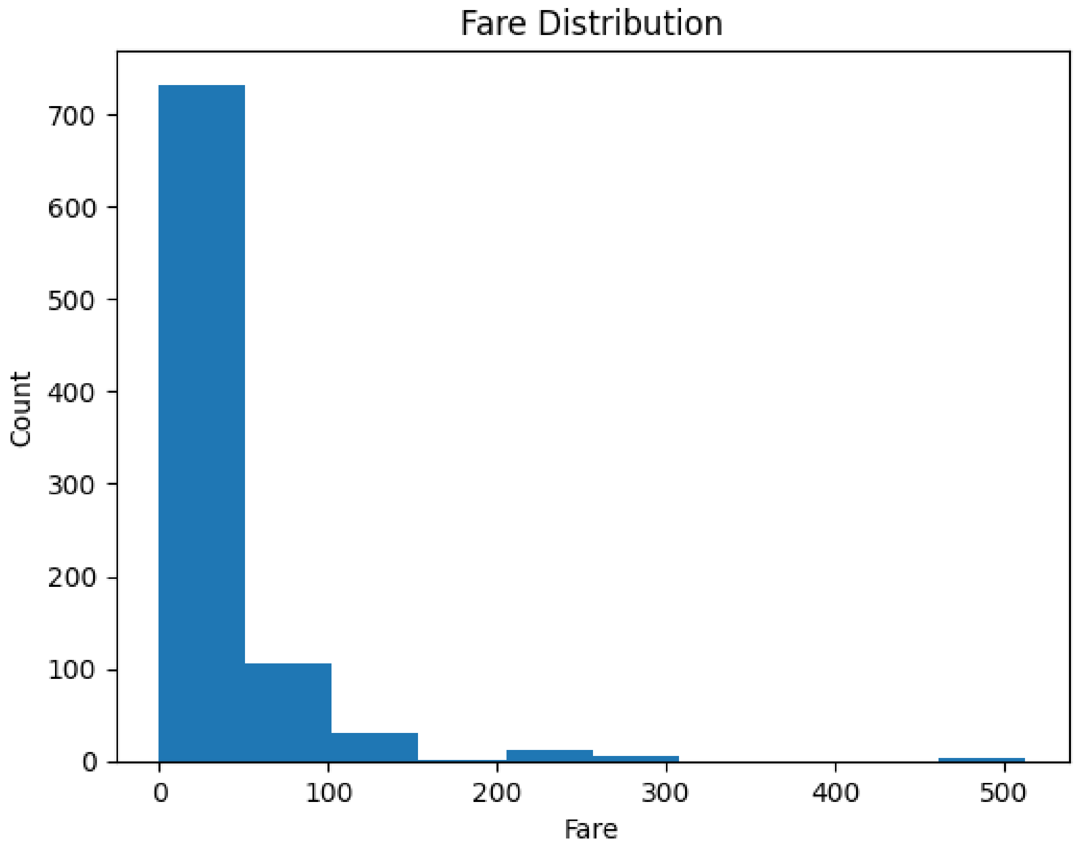
Fare 0

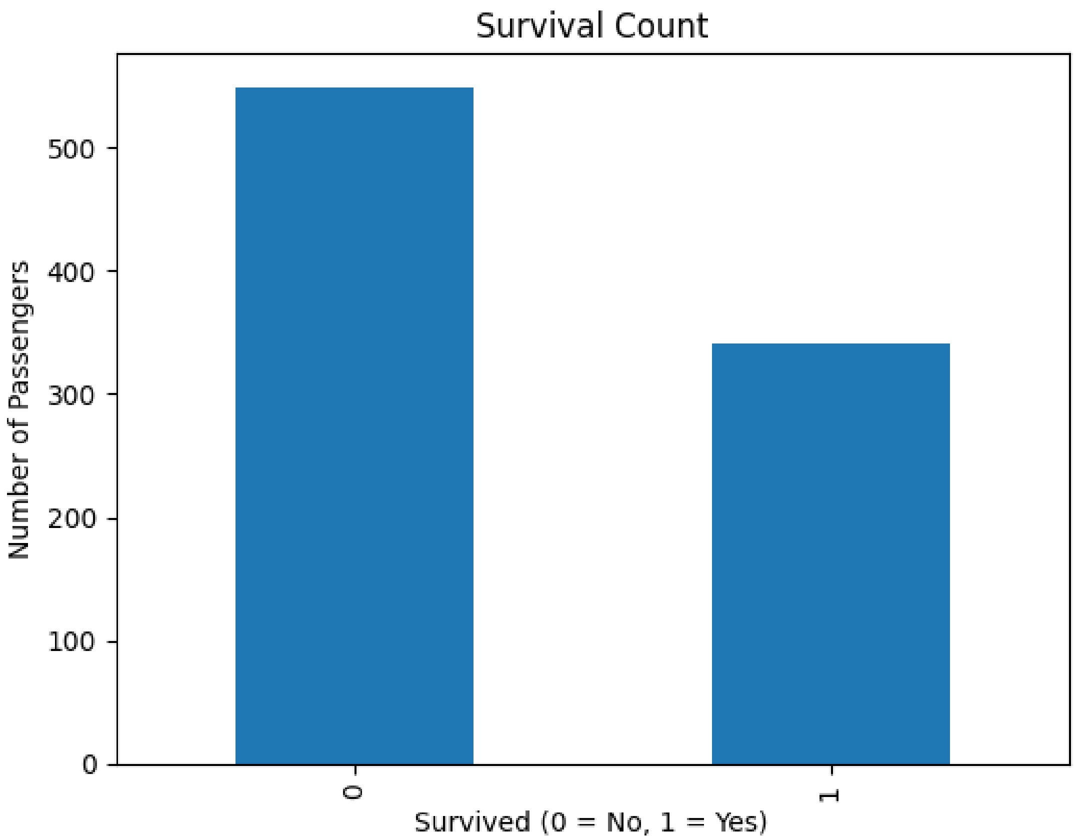
Cabin 687

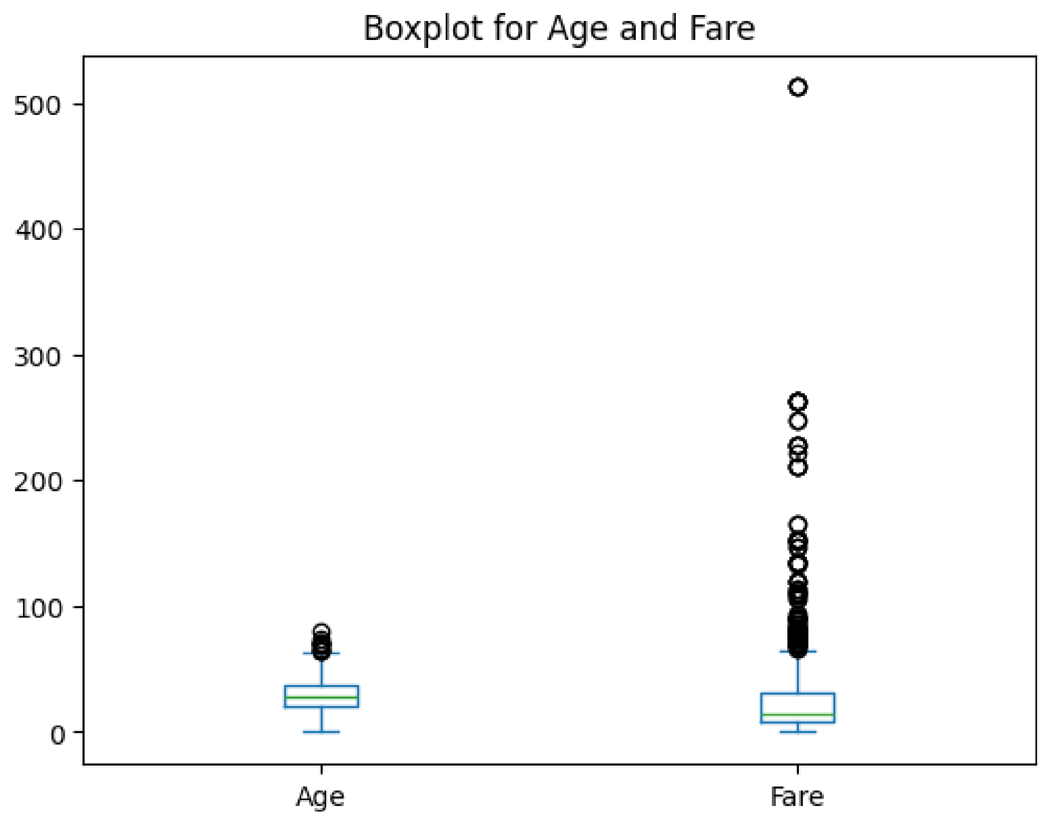
Embarked 2

dtype: int64




Fare Distribution
Count
700
600
500
400
300
200
100
0
0
100
200
300
400
500
Fare


Survival Count
Number of Passengers
500
400
300
200
100
0
0
1
Survived (0 = No, 1 = Yes)



Fare Outliers:

Fare

1 71.2833

27 263.0000

31 146.5208

34 82.1708

52 76.7292

Number of Fare Outliers: 116

Result

The dataset was analyzed to obtain basic statistical information and identify missing values. Visualizations such as histograms, bar plots, and box plots were used to understand the distribution of age, fare, and survival patterns. The IQR method was applied to detect outliers in the 'Fare' column, successfully identifying extreme values that deviate from the normal range.